Hybrid LSMC–PDE for a Generalized Gatheral Double Mean-Reverting Model

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Abstract

We study Bermudan put pricing under a generalized Gatheral double mean-reverting stochastic-volatility model. The model replaces the square-root volatility-of-volatility terms by Constant Elasticity of Variance (CEV)-type powers while preserving the double mean-reverting variance structure. The CEV terms create a boundary issue at zero: for exponents below one, the variance diffusion coefficients are not Lipschitz at the boundary, while the model is meaningful only if the variance factors remain nonnegative. We prove strong well-posedness and nonnegativity for elasticity exponents in $[\mathbf{1/2}, \mathbf{1}]$. We then derive the conditional representation needed to use the Hybrid Least-Squares Monte Carlo–Partial Differential Equation (LSMC–PDE) method in this model. The variance equations do not involve the asset price, so over one Bermudan step we condition on the Brownian increments driving the variance factors. After projecting the asset Brownian motion onto these drivers, at most one residual Brownian component remains. This gives a one-dimensional conditional equation in log-price space and an explicit Gaussian representation. These formulas provide the model-specific input for the hybrid regression convergence theorem and for the Fast Fourier Transform (FFT)-based implementation. Numerical experiments for Bermudan puts use a fixed parameter set based on a calibrated double mean-reverting specification from the literature. The Hybrid LSMC–PDE estimator is compared with plain Least-Squares Monte Carlo (LSMC) by varying both the Euler step count and the Monte Carlo path budget.

Keywords: Bermudan option pricing, stochastic volatility, Hybrid LSMC–PDE, least-squares Monte Carlo, generalized Gatheral model

1 Introduction

Bermudan option pricing under stochastic volatility requires repeated approximation of conditional continuation values. Deterministic Partial Differential Equation (PDE) methods handle this recursion well in low-dimensional Markovian models, but their cost grows quickly when several volatility factors are present. Simulation avoids a full state-space grid, but leaves a regression problem. In stochastic-volatility models that regression depends on both the asset level and the volatility state.

We study Bermudan put pricing under a generalized Gatheral-type double mean-reverting stochastic-volatility model. The instantaneous variance v mean-reverts to a stochastic variance level v', while v' mean-reverts to a long-run level. We replace the square-root volatility-of-volatility terms by Constant Elasticity of Variance (CEV)-type powers v^{δ_1} and $(v')^{\delta_2}$, with $\delta_1, \delta_2 \in [1/2, 1]$. The square-root model is recovered when $\delta_1 = \delta_2 = 1/2$.

The CEV generalization creates a boundary issue at zero. For exponents below one, the variance diffusion coefficients are not Lipschitz there, while the model is meaningful only as long as the variance factors remain nonnegative. We prove strong existence, pathwise uniqueness, and nonnegativity of the two-factor variance system. The proof uses the triangular structure of the model: first v', then v conditional on v', and finally the asset equation through its stochastic exponential.

The numerical method builds on the Hybrid Least-Squares Monte Carlo–PDE (LSMC–PDE) framework of Farahany, Jackson, and Jaimungal [1]. We use their convergence theorem as a general result and focus on the model-specific work needed for the Gatheral double mean-reverting (GDMR) dynamics. Since the variance equations do not involve the asset price, we condition on the Brownian increments driving the variance factors over one Bermudan step. After projecting the asset Brownian motion onto these drivers, at most one residual Brownian component remains, and the conditional asset-price step becomes a one-dimensional equation in log-price space. This conditioning is used only inside an iterated expectation and does not enlarge the exercise filtration.

The key quantities in this reduction are the correlated Brownian shift Z_n, the integrated drift I_n, the integrated residual variance B_n, and the terminal volatility state. These path statistics are the model-specific quantities needed to use the hybrid method. We then implement the method by combining Fast Fourier Transform (FFT) propagation in the asset direction with least-squares regression over volatility states.

The work is related to regression-based Bermudan pricing methods such as Least-Squares Monte Carlo (LSMC) [2–4], mixed Monte Carlo–PDE (MC–PDE) methods for stochastic-volatility models [1, 5–7], and stochastic-volatility models with mean-reverting or CEV-type variance dynamics [8–15].

The rest of the paper is organized as follows. Section 2 introduces the model, states the Bermudan recursion, and gives the well-posedness result. Section 3 derives the conditional hybrid representation. Section 4 describes the numerical implementation, Section 5 reports the experiments, and Section 6 concludes.

2 Gatheral double mean-reverting model

This section introduces the Gatheral double mean-reverting model (GDMR). In this model, the equations for the variance factors do not involve the asset price, while the asset equation depends on the variance factors. This one-way coupling is used in Section 3 to obtain the conditional one-dimensional representation.

2.1 GDMR dynamics

Fix a finite horizon $T > 0$. Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{Q})$$

be a filtered probability space satisfying the usual conditions. Throughout the paper, option pricing is performed under a risk-neutral (equivalent martingale) measure $\mathbb{Q}$ under which the discounted stock price process is assumed to be a true martingale. We assume a determinstic money-market account $B_t = e^{rt}$. Unless otherwise specified, all expectations are taken with respect to the probability measure $\mathbb{Q}$. Furthermore, the notation $\mathbb{E}_\xi[\cdot]$ denotes expectation with respect to the probability distribution of the random variable ξ.

The proability space supports a three-dimensional correlated $(\mathcal{F}_t)$-Brownian motion

$$W = (W^{(1)}, W^{(2)}, W^{(3)})$$

with constant quadratic covariations

$$\langle W^{(i)}, W^{(j)}\rangle_t = \rho_{ij} t, \qquad 1 \le i, j \le 3.$$

The matrix $\rho = (\rho_{ij})_{1\le i,j\le 3}$ is symmetric positive semidefinite and satisfies $\rho_{ii} = 1$. We assume $|\rho_{23}| < 1$. This ensures that the covariance matrix of the two variance Brownian drivers, $W^{(2)}$ and $W^{(3)}$, is nonsingular. The assumption is needed in Section 3 where $W^{(1)}$ is decomposed into its projection onto the span of $(W^{(2)}, W^{(3)})$ and an orthogonal residual Brownian component.

Let $S_0 > 0$, $v_0 \ge 0$, and $v'_0 \ge 0$. We consider the GDMR dynamics

$$\begin{aligned}
\mathrm{d}S_t &= r\, S_t \,\mathrm{d}t + S_t \sqrt{v_t}\, \mathrm{d}W_t^{(1)},\\
\mathrm{d}v_t &= \kappa_1\,(v'_t - v_t)\,\mathrm{d}t + \xi_1 v_t^{\delta_1}\, \mathrm{d}W_t^{(2)},\\
\mathrm{d}v'_t &= \kappa_2\,(\theta - v'_t)\,\mathrm{d}t + \xi_2 (v'_t)^{\delta_2}\, \mathrm{d}W_t^{(3)},
\end{aligned} \tag{1}$$

where $r \ge 0$, $\kappa_1, \kappa_2, \theta, \xi_1, \xi_2 > 0$, and $\delta_1, \delta_2 \in [1/2, 1]$. The factor v is the instantaneous variance and mean-reverts to the stochastic level v', while v' mean-reverts to the long-run level θ. The exponents δ_1 and δ_2 replace the square-root volatility-of-volatility coefficients by CEV-type powers; the square-root double mean-reverting model is recovered when $\delta_1 = \delta_2 = 1/2$.

We restrict further the leverage correlations to satisfy

$$\rho_{12} < 0, \qquad \rho_{13} < 0.$$

In particular, negative stock–volatility correlations are known to mitigate potential moment explosion effects and support stable martingale behavior in stochastic volatility models as pointed out in Andersen and Piterbarg [16]. This assumption is also consistent with market stylized facts on stock–volatility correlations.

2.2 Well-posedness and nonnegativity

The model is formulated on the nonnegative variance state space

$$\mathcal{D} := (0, \infty) \times [0, \infty)^2.$$

We next verify that this state space is preserved. The only coefficients that may fail to be Lipschitz are the CEV-type variance diffusions. For the proof of strong well-posedness, we extend only the variance diffusion coefficients to the real line by setting

$$\sigma_i(x) := \xi_i (x^+)^{\delta_i}, \qquad x^+ := \max\{x, 0\}, \qquad i = 1, 2.$$

Once Theorem 2.2 is proved, the variance factors remain nonnegative, and therefore the extended coefficients coincide with the original coefficients along the solution paths of (1).

The well-posedness and boundary behavior of the Gatheral double mean-reverting volatility system have recently been discussed by Mishura, Pilipenko, and Ralchenko [15]. They prove strong existence, uniqueness, nonnegativity, and the strong Markov property for a closely related double-CEV Gatheral system, and also study the possibility that the volatility factors spend time arbitrarily close to zero. Our use of this result is narrower: we keep the original, non-reflected GDMR model and need only strong well-posedness and nonnegativity for the pricing construction. For this reason, the proof of Theorem 2.2 is written in the form needed here. It uses the triangular structure of (1): first solve the v'-equation, then solve the v-equation with v' as an adapted input, and finally recover the asset equation from its stochastic exponential. The same one-way structure is used later in Subsection 3.2 and Proposition 3.1 when conditioning on the variance Brownian increments in the Hybrid LSMC–PDE step.

Assumption 2.1 (CEV exponents) Throughout this section we impose

$$\delta_1, \delta_2 \in [1/2, 1].$$

For $\delta \in [1/2, 1]$, the map $x \mapsto (x^+)^\delta$ satisfies

$$\left|(x^+)^\delta - (y^+)^\delta\right| \le |x - y|^\delta, \qquad x, y \in \mathbb{R}.$$

Hence the diffusion coefficient $x \mapsto \xi(x^+)^\delta$ satisfies the Yamada–Watanabe modulus condition [14]. In the one-dimensional form used here, this condition means that there exists a nondecreasing function ϱ with $\varrho(0) = 0$ and $\varrho(u) > 0$ for $u > 0$ such that

$$|\sigma(x) - \sigma(y)| \le \varrho(|x - y|), \qquad x, y \in \mathbb{R},$$

and

$$\int_0^{\varepsilon} \frac{\mathrm{d}u}{\varrho(u)^2} = \infty$$

for every sufficiently small $\varepsilon > 0$. For $\sigma(x) = \xi(x^+)^\delta$, the previous inequality gives this condition with $\varrho(u) = Cu^\delta$, since

$$\int_0^{\varepsilon} u^{-2\delta}\,\mathrm{d}u = \infty \qquad \text{for } \delta \geq \frac{1}{2}.$$

The upper bound $\delta \leq 1$ gives at most linear growth.

Theorem 2.2 (Strong well-posedness and nonnegativity) *Assume Assumption 2.1. For every $T > 0$ and every initial state*

$$S_0 > 0, \qquad v_0 \geq 0, \qquad v_0' \geq 0,$$

the nonnegative-state system (1) admits a unique strong solution $(S_t, v_t, v_t')_{t\in[0,T]}$ with continuous paths such that

$$S_t > 0, \qquad v_t \geq 0, \qquad v_t' \geq 0, \qquad 0 \leq t \leq T, \quad \mathbb{Q}\text{-}a.s.$$

Moreover, for every deterministic initial state in $\mathcal{D}$, the solution is a time-homogeneous strong Markov process on $\mathcal{D}$.

The proof is given in Appendix A. It uses the triangular structure of the variance equations: solve the v'-equation first, then the v-equation with v' as an adapted input, and finally recover the asset from its stochastic exponential. The positive-part extension and the one-dimensional invariance criterion keep both variance factors nonnegative.

The lower bound $\delta_i \geq 1/2$ is the Yamada–Watanabe threshold for pathwise uniqueness of power diffusion coefficients. We do not impose a Feller-type condition, since the theorem proves nonnegativity rather than strict positivity or boundary inaccessibility.

3 Bermudan option pricing

We now pass from the model dynamics to the one-step continuation calculation in the Bermudan recursion. The purpose of this section is to explain how the Hybrid LSMC–PDE method is used for the GDMR model and why the Brownian projection is needed. The Hybrid LSMC–PDE method was developed by Farahany, Jackson, and Jaimungal [1] for multidimensional stochastic volatility models. In this paper, our contribution is to adapt that general framework to the GDMR model by deriving the Brownian projection and the resulting one-dimensional conditional log-price representation. We use the general hybrid-regression convergence theorem of Farahany et al. [1]; the representation derived here provides the model-specific input needed to apply that theorem.

3.1 Bermudan stopping problem

Let

$$\pi = \{0 = t_0 < t_1 < \cdots < t_M = T\}$$

be the Bermudan exercise grid. At t_n, the payoff is $h_n(S_{t_n})$, where $h_n : (0,\infty) \to \mathbb{R}$ is Borel. We assume

$$\mathbb{E}\left[\max_{0\le n\le M} e^{-rt_n}|h_n(S_{t_n})|\right] < \infty.$$

When the risk-free interest rate is nonnegative, early exercise of a Bermudan call option on a non-dividend-paying stock is not optimal. Therefore, we focus on Bermudan put options, with payoff $h_n(s) = (K-s)^+$.

By Theorem 2.2, the state process (S_t, v_t, v'_t) is a time-homogeneous Markov process on

$$\mathcal{D} = (0,\infty) \times [0,\infty)^2.$$

Thus the Bermudan value functions may be defined by backward induction. Write

$$\Delta t_n := t_{n+1} - t_n$$

and set

$$U_M(s,v,w) := h_M(s),$$

where w denotes the second variance coordinate. For $n = M-1, \ldots, 0$, define

$$C_n(s,v,w) := e^{-r\Delta t_n}\mathbb{E}\left[U_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \,\middle|\, S_{t_n} = s,\ v_{t_n} = v,\ v'_{t_n} = w\right],$$

and

$$U_n(s,v,w) := \max\{h_n(s), C_n(s,v,w)\}.$$

The time-zero Bermudan value is

$$V_0 = U_0(S_0, v_0, v'_0).$$

3.2 Hybrid decomposition of the continuation value

A plain LSMC method approximates the continuation value in the Bermudan recursion by regressing simulated discounted next-step values on the current state. In a stochastic-volatility model this regression is more demanding than in a one-factor asset model, because the continuation value depends on both the asset level and the variance state. The Hybrid LSMC–PDE method reduces this regression problem by using the one-way structure of the dynamics. The variance factors are simulated first; along each simulated variance path, the remaining conditional expectation in the asset direction is computed by a one-dimensional PDE step; the resulting pathwise conditional values are then regressed over the current variance state. The numerical realization of these steps, including the FFT evaluation and the regression matrices, is given in Section 4.

The idea can be written directly at the level of one Bermudan interval. Let F be a next-step value function; in the backward recursion, F is U_{n+1} or its numerical approximation. For a current state (s, v, w), define

$$C_n^F(s,v,w) = e^{-r\Delta t_n}\mathbb{E}\Big[F(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \,\Big|\, S_{t_n} = s,\ v_{t_n} = v,\ v'_{t_n} = w\Big].$$

The hybrid method evaluates this conditional expectation in two stages, using the one-way coupling of the model. The equations for (v, v') do not involve S, so over one exercise interval we may first simulate the Brownian increments driving the variance factors. Conditioning on these increments fixes the variance segment and the part of the asset Brownian motion correlated with the variance drivers. The only remaining asset randomness is the orthogonal Brownian component. This conditioning is used only inside the one-step conditional expectation and does not change the information used for the exercise decision at the exercise dates.

Let $\mathcal{H}_n$ denote this enlarged conditioning information; its precise definition is given in the next subsection. The inner, pathwise continuation value is

$$\bar{C}_n^F(s;\mathcal{H}_n) := e^{-r\Delta t_n}\mathbb{E}\Big[F(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \,\Big|\, S_{t_n} = s, \mathcal{H}_n\Big].$$

For each simulated variance path segment, $\bar{C}_n^F(\cdot;\mathcal{H}_n)$ is a function of the initial asset level. The original continuation value is recovered by the outer conditional expectation

$$C_n^F(s,v,w) = \mathbb{E}\big[\bar{C}_n^F(s;\mathcal{H}_n) \,\big|\, v_{t_n} = v,\ v'_{t_n} = w\big].$$

This expectation is the part approximated by least-squares regression over the current variance state.

To compute the conditional value, we must isolate the part of the asset Brownian motion that remains random after conditioning on $\mathcal{H}_n$. This is the role of the Brownian projection below. Once this residual component is identified, the conditional asset-price problem becomes one-dimensional in the log-price variable. The conditioning is only an iterated-expectation device and does not change the exercise information at time t_n.

3.3 Brownian projection

We now make the conditioning information in the preceding decomposition explicit. Fix an interval $[t_n, t_{n+1}]$ in the Bermudan exercise grid and define

$$\mathcal{H}_n := \mathcal{F}_{t_n} \vee \sigma\Big(W_s^{(2)} - W_{t_n}^{(2)}, W_s^{(3)} - W_{t_n}^{(3)} : s \in [t_n, t_{n+1}]\Big).$$

Since v and v' are driven only by $W^{(2)}$ and $W^{(3)}$, the path $(v_s, v'_s)_{s\in[t_n,t_{n+1}]}$ is $\mathcal{H}_n$-measurable. We project $W^{(1)}$ onto the span of $(W^{(2)}, W^{(3)})$. Set

$$\Sigma_{23} := \begin{bmatrix} 1 & \rho_{23} \\ \rho_{23} & 1 \end{bmatrix}, \qquad \beta := \Sigma_{23}^{-1}\begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix}.$$

Equivalently,

$$\beta_2 = \frac{\rho_{12} - \rho_{13}\rho_{23}}{1 - \rho_{23}^2}, \qquad \beta_3 = \frac{\rho_{13} - \rho_{12}\rho_{23}}{1 - \rho_{23}^2}. \tag{2}$$

The residual variance is

$$\begin{aligned} \sigma_\perp^2 &:= 1 - \begin{bmatrix} \rho_{12} & \rho_{13} \end{bmatrix} \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} \\ &= \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1 - \rho_{23}^2} \geq 0. \end{aligned} \tag{3}$$

When $\sigma_\perp > 0$,

$$B_t^\perp := \frac{W_t^{(1)} - \beta_2 W_t^{(2)} - \beta_3 W_t^{(3)}}{\sigma_\perp}$$

is a Brownian motion, and its increments over $[t_n, t_{n+1}]$ are independent of $\mathcal{H}_n$. Hence

$$\mathrm{d}W_t^{(1)} = \beta_2 \,\mathrm{d}W_t^{(2)} + \beta_3 \,\mathrm{d}W_t^{(3)} + \sigma_\perp \,\mathrm{d}B_t^\perp. \tag{4}$$

If $\sigma_\perp = 0$, the residual term is absent and the asset Brownian driver is fully determined by the Brownian drivers of the variance factors.

3.4 Conditional one-dimensional equation

Let $X_t := \log S_t$. Substituting (4) into the asset equation gives

$$\mathrm{d}X_t = \Big(r - \tfrac{1}{2}v_t\Big)\mathrm{d}t + \sqrt{v_t}\Big(\beta_2 \,\mathrm{d}W_t^{(2)} + \beta_3 \,\mathrm{d}W_t^{(3)}\Big) + \sigma_\perp \sqrt{v_t}\,\mathrm{d}B_t^\perp.$$

Define the $\mathcal{H}_n$-measurable shift

$$Z_n := \int_{t_n}^{t_{n+1}} \sqrt{v_s}\Big(\beta_2 \,\mathrm{d}W_s^{(2)} + \beta_3 \,\mathrm{d}W_s^{(3)}\Big).$$

For $t \in [t_n, t_{n+1}]$, define also

$$I_n(t) := \int_t^{t_{n+1}} \Big(r - \tfrac{1}{2}v_s\Big)\mathrm{d}s, \qquad B_n(t) := \sigma_\perp^2 \int_t^{t_{n+1}} v_s \,\mathrm{d}s,$$

and write $I_n := I_n(t_n)$ and $B_n := B_n(t_n)$.

Proposition 3.1 (Conditional one-step representation) *Let $G : \mathbb{R} \times [0, \infty)^2 \to \mathbb{R}$ be bounded and Borel. Conditionally on $\mathcal{H}_n$, the function*

$$u(t, y) := \mathbb{E}\Big[G\Big(Y_{t_{n+1}}^{t,y} + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}\Big) \,\Big|\, \mathcal{H}_n\Big],$$

where

$$\mathrm{d}Y_s^{t,y} = \Big(r - \tfrac{1}{2}v_s\Big)\mathrm{d}s + \sigma_\perp \sqrt{v_s}\,\mathrm{d}B_s^\perp, \qquad Y_t^{t,y} = y,$$

is the bounded mild solution of

$$\partial_t u + \Big(r - \tfrac{1}{2}v_t\Big)\partial_y u + \frac{1}{2}\sigma_\perp^2 v_t \partial_{yy} u = 0, \qquad u(t_{n+1}, y) = G(y + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}).$$

Equivalently, for an auxiliary $\xi \sim N(0,1)$,

$$u(t,y) = \mathbb{E}_\xi\Big[G\Big(y + Z_n + I_n(t) + \sqrt{B_n(t)}\,\xi, v_{t_{n+1}}, v'_{t_{n+1}}\Big)\Big], \tag{5}$$

with the deterministic interpretation when $B_n(t) = 0$.

Proof. After conditioning on $\mathcal{H}_n$, the variance path and the shift Z_n are fixed. If $\sigma_\perp > 0$, the only remaining randomness is the residual Brownian motion; if $\sigma_\perp = 0$, the conditional step is deterministic. The conditional Feynman–Kac formula gives the backward Kolmogorov equation. Since the coefficients are independent of y, the terminal log-price is Gaussian with mean $y + I_n(t)$ and variance $B_n(t)$, which gives the displayed formula.

For an asset-space next-step value function F and $G(y,v,w) := F(e^y, v, w)$, the pathwise quantity defined in Subsection 3.2 is

$$\bar{C}_n^F(s; \mathcal{H}_n) = e^{-r\Delta t_n} u(t_n, \log s).$$

Thus the conditional asset step depends on the sampled $(W^{(2)}, W^{(3)})$-path over $[t_n, t_{n+1}]$ only through

$$Z_n, \qquad I_n, \qquad B_n, \qquad v_{t_{n+1}}, \qquad v'_{t_{n+1}},$$

while (v_{t_n}, v'_{t_n}) is the state used in the outer regression.

3.5 Connection with the hybrid convergence theorem

The convergence theorem for the truncated Hybrid LSMC–PDE regression is proved in [1]. The preceding subsections identify the model-specific ingredients needed to apply that theorem.

First, the asset equation is multiplicative. For $0 \le t < u \le T$,

$$S_u = S_t R_{t,u},$$

where

$$R_{t,u} = \exp\left(\int_t^u \left(r - \frac{1}{2}v_s\right) \mathrm{d}s + \int_t^u \sqrt{v_s}\, \mathrm{d}W_s^{(1)}\right).$$

Thus the return factor $R_{t,u}$ depends on the Brownian and variance paths, but not on the value of S_t.

Second, Proposition 3.1 shows that, on each Bermudan interval, the sampled $(W^{(2)}, W^{(3)})$-path enters the conditional asset-price step only through the finite-dimensional statistic

$$\Theta_n := (V_n, V_{n+1}, Z_n, I_n, B_n), \qquad V_n = (v_{t_n}, v'_{t_n}).$$

Because the variance equations do not involve S, the model is one-way coupled. Hence we can apply the hybrid convergence theorem of [1, Theorem 1].

With the compact volatility truncation, bounded basis functions, and nonsingular population Gram matrices required in [1], the stabilized hybrid regression coefficients converge almost surely. For each fixed asset level, the fitted continuation values converge uniformly on the chosen compact volatility domain.

4 Numerical implementation

The numerical experiments use the following implementation. The conditional representation is derived in Section 3; here we specify the grids, variance discretization, accumulated path statistics, FFT step, regression step, and time-zero estimator.

4.1 Grids and basis functions

Let

$$\pi = \{0 = t_0 < t_1 < \cdots < t_M = T\}$$

be the Bermudan exercise grid, and let h_n denote the exercise payoff at t_n. We work on a uniform log-price grid

$$y_i = y_{\min} + (i-1)\Delta y, \qquad \Delta y = \frac{y_{\max} - y_{\min}}{N_S - 1}, \qquad i = 1, \ldots, N_S,$$

and set

$$s_i = e^{y_i}, \qquad \mathcal{S}_h := \{s_1, \ldots, s_{N_S}\}.$$

The interval $[y_{\min}, y_{\max}]$ is chosen large enough to contain the relevant asset values used in the computation.

For the volatility variables, write $w = v'$. Choose a compact rectangle

$$\mathcal{D}_v = [0, v_{\max}] \times [0, w_{\max}] \subset [0, \infty)^2$$

and a vector of continuous bounded basis functions

$$\phi = (\phi_1, \ldots, \phi_{d_B}), \qquad \operatorname{supp}(\phi_k) \subset \mathcal{D}_v, k = 1, \ldots, d_B.$$

Once chosen, the domain $\mathcal{D}_v$ and the basis functions are fixed before the training regression.

At each exercise date, we approximate the continuation value by a separable regression form

$$c_n(s, v, w) \approx a_n(s) \cdot \phi(v, w), \qquad a_n(s) \in \mathbb{R}^{d_B}.$$

Numerically, the coefficient vectors $a_n(s)$ are computed only at the grid points $s_i \in \mathcal{S}_h$. Off-grid asset values are evaluated by interpolation in the log-price coordinate.

4.2 Variance simulation and path statistics

Let

$$0 = \tau_0 < \tau_1 < \cdots < \tau_L = T$$

be an Euler simulation grid containing all exercise dates in the Bermudan exercise grid π. Write

$$\Delta\tau_\ell := \tau_{\ell+1} - \tau_\ell.$$

For each path j, simulate correlated Gaussian increments

$$(\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j})$$

with variances $\Delta\tau_\ell$ and covariance $\rho_{23}\Delta\tau_\ell$. The numerical implementation uses the following positive-part Euler discretization:

$$\begin{aligned}
\bar{w}_\ell^j &:= (w_\ell^j)^+, \qquad \bar{v}_\ell^j := (v_\ell^j)^+, \\
w_{\ell+1}^j &:= \left[w_\ell^j + \kappa_2(\theta - \bar{w}_\ell^j)\Delta\tau_\ell + \xi_2(\bar{w}_\ell^j)^{\delta_2}\Delta W_\ell^{(3),j}\right]^+, \\
v_{\ell+1}^j &:= \left[v_\ell^j + \kappa_1(\bar{w}_\ell^j - \bar{v}_\ell^j)\Delta\tau_\ell + \xi_1(\bar{v}_\ell^j)^{\delta_1}\Delta W_\ell^{(2),j}\right]^+.
\end{aligned}$$

For each exercise interval $[t_n, t_{n+1}]$, the path statistics are accumulated over the Euler substeps contained in that interval:

$$\begin{aligned}
Z_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \sqrt{\bar{v}_\ell^j}\left(\beta_2\Delta W_\ell^{(2),j} + \beta_3\Delta W_\ell^{(3),j}\right), \\
I_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \left(r - \frac{1}{2}\bar{v}_\ell^j\right)\Delta\tau_\ell, \\
B_n^j &:= \sigma_\perp^2 \sum_{\tau_\ell \in [t_n, t_{n+1})} \bar{v}_\ell^j\Delta\tau_\ell.
\end{aligned}$$

Here the projection coefficients β_2, β_3 and the residual volatility coefficient $\sigma_\perp$ are given by Equations (2) and (3), respectively. The left-point rule is natural for Z_n^j, since Z_n^j approximates a stochastic integral evaluated with left endpoints. The same left-point values are used for I_n^j and B_n^j for consistency with the Euler simulation, and the same increments $\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j}$ are used in the variance updates and in Z_n^j.

4.3 Pathwise FFT propagation and regression

Fix $n \in \{0, \ldots, M-1\}$. Here $\widehat{V}_{n+1}^N$ denotes the numerical Bermudan value surface already available from the next exercise date in the backward recursion. At maturity this surface is initialized by $\widehat{V}_M^N(s_i, v, w) = h_M(s_i)$. At earlier exercise dates it is the previously updated surface

$$\widehat{V}_{n+1}^N(s_i, v, w) = \max\{h_{n+1}(s_i), \widehat{C}_{n+1}^N(s_i, v, w)\}, \qquad \widehat{C}_{n+1}^N(s_i, v, w) = a_{n+1}^N(s_i)\cdot\phi(v, w).$$

Thus $\widehat{V}_{n+1}^N$ is known on $\mathcal{S}_h \times \mathcal{D}_v$ before the following pathwise propagation step. For each Monte Carlo path j, the simulated Brownian increments driving the variance factors on $[t_n, t_{n+1}]$ determine the endpoint state

$$(v_{t_{n+1}}^j, w_{t_{n+1}}^j)$$

and the statistics Z_n^j, I_n^j, B_n^j. Define the terminal grid function

$$g_{n+1}^j(y_i) := \widehat{V}_{n+1}^N\left(e^{y_i}, v_{t_{n+1}}^j, w_{t_{n+1}}^j\right), \qquad i = 1, \ldots, N_S.$$

By the Gaussian representation (5) in Section 3,

$$u_n^j(y) = \mathbb{E}_\xi\left[g_{n+1}^j\left(y + Z_n^j + I_n^j + \sqrt{B_n^j}\xi\right)\right], \qquad \xi \sim N(0,1).$$

With the Fourier convention

$$\mathcal{F}f(\omega) = \int_{\mathbb{R}} e^{-\mathrm{i}\omega y} f(y)\,\mathrm{d}y,$$

we obtain

$$\mathcal{F}u_n^j(\omega) = \mathcal{F}g_{n+1}^j(\omega)\exp\left(\mathrm{i}\omega Z_n^j + \mathrm{i}\omega I_n^j - \frac{1}{2}\omega^2 B_n^j\right).$$

Define

$$\Psi_n^j(\omega) := \mathrm{i}\omega Z_n^j + \mathrm{i}\omega I_n^j - \frac{1}{2}\omega^2 B_n^j.$$

On the uniform log-price grid, the FFT implementation is

$$u_n^j(\cdot) \approx \mathrm{FFT}^{-1}\left(\mathrm{FFT}[g_{n+1}^j]\,\exp(\Psi_n^j)\right), \tag{6}$$

where the exponential is applied pointwise at the discrete Fourier frequencies. The FFT input is the unshifted terminal grid g_{n+1}^j; the correlated shift Z_n^j is included only through the multiplier. Equivalently, one may shift the terminal grid first and omit $\mathrm{i}\omega Z_n^j$, but the two conventions must not be combined. If $B_n^j = 0$, the conditional propagation reduces to the deterministic log-price shift $Z_n^j + I_n^j$.

The pathwise pre-surface on the asset grid is

$$\widehat{C}_n^j(s_i) := e^{-r\Delta t_n} u_n^j(y_i), \qquad i = 1, \ldots, N_S.$$

The regression step is performed at each exercise date t_n, $n = M-1, \ldots, 1$, and at each asset grid point s_i. It estimates the dependence of the pathwise pre-surface $\widehat{C}_n^j(s_i)$ on the current volatility state. Define

$$V_n^j := (v_{t_n}^j, w_{t_n}^j).$$

The sample Gram matrix is

$$A_n^N := \frac{1}{N}\sum_{j=1}^{N} \phi(V_n^j)\phi(V_n^j)^\top,$$

and the right-hand side for the grid point s_i is

$$b_n^N(s_i) := \frac{1}{N}\sum_{j=1}^{N} \phi(V_n^j)\widehat{C}_n^j(s_i).$$

For numerical stability, we use the truncated inverse

$$[A]_R^{-1} := \begin{cases} A^{-1}, & \text{if } A \text{ is invertible and } \|A^{-1}\| \le R, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Then set

$$a_n^N(s_i) := [A_n^N]_R^{-1} b_n^N(s_i). \tag{8}$$

The fitted continuation surface is

$$\widehat{C}_n^N(s_i, v, w) := a_n^N(s_i) \cdot \phi(v, w),$$

and the updated price for the Bermudan option is

$$\widehat{V}_n^N(s_i, v, w) := \max\left\{h_n(s_i), \widehat{C}_n^N(s_i, v, w)\right\}.$$

Starting from

$$\widehat{V}_M^N(s_i, v, w) := h_M(s_i),$$

the recursion is performed backward for $n = M-1, \dots, 1$. The fitted functions are defined at the asset grid points and extended to off-grid asset values by log-price interpolation.

4.4 Time-zero estimator and pseudocode

At $t_0 = 0$, the initial volatility state is fixed, so no regression is needed. After $\widehat{V}_1^N$ has been fitted, the time-zero continuation value is evaluated on the asset grid by averaging first-step pathwise pre-surfaces:

$$\widehat{C}_0^N(s_i) := \frac{1}{N}\sum_{j=1}^{N} \widehat{C}_0^j(s_i).$$

The averaged grid values define $\widehat{C}_0^N$ on the asset grid; off-grid values are evaluated by the log-price interpolation convention stated above. Thus $\widehat{C}_0^N(S_0)$ denotes the averaged continuation value at the initial asset price, and the Hybrid LSMC–PDE estimator is

$$\widehat{V}_0^{N,\mathrm{hyb}}(S_0) = \max\left\{h_0(S_0), \widehat{C}_0^N(S_0)\right\}. \tag{9}$$

Algorithm 1 Hybrid LSMC–PDE recursion

Require: Exercise grid $\{t_n\}_{n=0}^{M}$, Euler grid containing the exercise dates, log-price grid $\{y_i\}_{i=1}^{N_S}$, volatility domain $\mathcal{D}_v$, basis ϕ, truncation level R, and path count N

Ensure: Hybrid LSMC–PDE estimator $\widehat{V}_0^{N,\mathrm{hyb}}(S_0)$ and fitted continuation surfaces $\widehat{C}_n^N$

Simulate N training paths of the variance factors on the Euler grid.
Store exercise-date volatility states and accumulate Z_n^j, I_n^j, and B_n^j over each interval $[t_n, t_{n+1}]$.
Set $\widehat{V}_M^N(s_i, v, w) \leftarrow h_M(s_i)$ on $\mathcal{S}_h \times \mathcal{D}_v$.
for $n = M-1, \dots, 1$ **do**
for $j = 1, \dots, N$ **do**
Compute $\{\widehat{C}_n^j(s_i)\}_{i=1}^{N_S}$ using the unshifted terminal grid g_{n+1}^j in (6), with Z_n^j included only through the multiplier $\exp(\Psi_n^j)$; do not pre-shift g_{n+1}^j by Z_n^j.
end for
for each $s_i \in \mathcal{S}_h$ **do**
Compute $b_n^N(s_i)$ and $a_n^N(s_i)$ by the truncated-inverse regression (8).
end for
Set $\widehat{C}_n^N(s_i, v, w) \leftarrow a_n^N(s_i) \cdot \phi(v, w)$ for all $s_i \in \mathcal{S}_h$.
Set $\widehat{V}_n^N(s_i, v, w) \leftarrow \max\{h_n(s_i), \widehat{C}_n^N(s_i, v, w)\}$ for all $s_i \in \mathcal{S}_h$.
end for
Compute the first-step pre-surfaces $\widehat{C}_0^j(s_i)$ for $j = 1, \dots, N$ using the simulated paths on $[t_0, t_1]$.
Set $\widehat{C}_0^N(s_i) \leftarrow N^{-1} \sum_{j=1}^{N} \widehat{C}_0^j(s_i)$.
Evaluate $\widehat{C}_0^N(S_0)$ from the averaged continuation grid at $y = \log S_0$, using linear interpolation if needed.
return $\widehat{V}_0^{N,\mathrm{hyb}}(S_0) = \max\{h_0(S_0), \widehat{C}_0^N(S_0)\}$.

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5 Numerical experiments

We report Bermudan put pricing experiments for the fixed parameter set in Table 1. Since no closed-form solution is available for Bermudan option prices under the GDMR model, we compare a plain LSMC estimator with our Hybrid LSMC–PDE estimator, using large-simulation plain LSMC estimates as reference prices. The plain LSMC estimator [2] and the Hybrid LSMC–PDE estimator in Equation (9) are compared using reference-relative errors. Standard errors and 95% confidence intervals are reported for the reference prices and used to construct the error bars in the figures.

5.1 Experimental setting and reference values

Table 1 reports the fixed GDMR parameter set and numerical settings used in the experiments.

Table 1 Parameter set and numerical settings used in the Bermudan pricing experiments.

Symbol	Value	Interpretation
S_0	100	initial asset price
v_0	0.114	initial instantaneous variance factor
v'_0	0.110	initial secondary variance factor
r	0.02	risk-free rate
T	1	maturity in years
κ_1	5.5	mean-reversion speed of v_t toward v'_t
κ_2	0.1	mean-reversion speed of v'_t toward θ
θ	0.078	long-run variance level
ξ_1	2.689	volatility scale of the first variance factor
ξ_2	0.502	volatility scale of the second variance factor
δ_1	0.94	CEV exponent of the first variance factor
δ_2	0.94	CEV exponent of the second variance factor
ρ_{12}	-0.982	correlation between the first and second Brownian drivers
ρ_{13}	-0.727	correlation between the first and third Brownian drivers
ρ_{23}	0.590	correlation between the second and third Brownian drivers
Fixed numerical setting		
N_{ex}	12	exercise dates after t_0, including maturity
K	$\{70, 80, 90, 100, 110\}$	strike set used in all experiments
N_S	301	asset-grid points in the conditional PDE solver
$(a_{\min}, a_{\max})$	$(0.30, 3.50)$	asset-range factors relative to S_0
q_v	0.999	volatility truncation quantile
$(M_{\mathrm{ref}}, N_{\mathrm{ref}})$	(1200, 1,200,000)	reference LSMC step count and path budget

The LSMC reference prices are reported in Table 2. The Hybrid LSMC–PDE grid and truncation settings are kept fixed throughout both sets of experiments.

The mean-reversion, volatility-scale, initial-variance, and correlation parameters are taken from the calibrated double mean-reverting specification of Bayer, Gatheral and Karlsmark [11]. Their calibration uses $r = 0$. In the present Bermudan pricing experiments we keep those variance and correlation parameters, set $r = 0.02$, and use $\delta_1 = \delta_2 = 0.94$ for the CEV-type generalization. Thus the numerical section is not a new calibration study; it uses a parameter set from the double mean-reverting volatility literature as a fixed test case. The correlation assumptions used in Section 3 are satisfied. In particular, $|\rho_{23}| < 1$.

The plain LSMC regressions use a cubic 16-term basis

$$1, x, y, z, x^2, y^2, z^2, xy, xz, yz, x^3, y^3, z^3, p, p^2, p^3,$$

where $x = S/S_0$, $y = v/\theta$, $z = v'/\theta$, and $p = (K - S)^+/K$. The hybrid regressions use a volatility-only cubic basis

$$1, y, z, y^2, yz, z^2, y^3, y^2z, yz^2, z^3,$$

where y and z are the normalized volatility coordinates after applying the numerical truncation to the chosen volatility rectangle D_v. The smooth compact-support assumption in Subsection 3.5 is part of the convergence statement.

For an estimator V and reference value V^{ref}, we report

$$\varepsilon_{\mathrm{rel}} := \frac{|V - V^{\mathrm{ref}}|}{|V^{\mathrm{ref}}|}.$$

The reference values are plain LSMC estimates computed with 1200 Euler steps and 1,200,000 simulation paths. They are used as numerical reference values, not as exact prices. The reported standard errors are empirical Monte Carlo standard errors of the estimators, and the 95% confidence intervals are computed as the estimate plus or minus 1.96 standard errors. In the relative-error plots, the reference value V^{ref} is treated as fixed. The error bars are obtained by mapping the estimator confidence intervals through

$$V \mapsto \frac{|V - V^{\mathrm{ref}}|}{|V^{\mathrm{ref}}|}.$$

Table 2 LSMC reference values for the parameter set in Table 1.

K	Reference price	Std. error	95% confidence interval
70	2.522851	0.007103	[2.508929, 2.536773]
80	4.298933	0.009345	[4.280617, 4.317249]
90	6.942119	0.011774	[6.919042, 6.965196]
100	10.682037	0.014105	[10.654391, 10.709683]
110	15.743520	0.015757	[15.712636, 15.774404]

5.2 Fixed path budget and varying Euler steps

The first experiment keeps the Monte Carlo path budget fixed and varies the number of Euler time steps. The two path budgets are 20,000 and 60,000, and both methods are evaluated at $24, 48, 72$, and 96 Euler steps. This experiment tests the sensitivity of the estimators to the Euler discretization.

For 20,000 paths, the LSMC reference-error range over the full step-sweep grid is 0.344%–8.645%, while the Hybrid LSMC–PDE range is 0.019%–7.459%. Figure 1 reports the strike-wise reference errors. The corresponding price grids are reported in Appendix B.

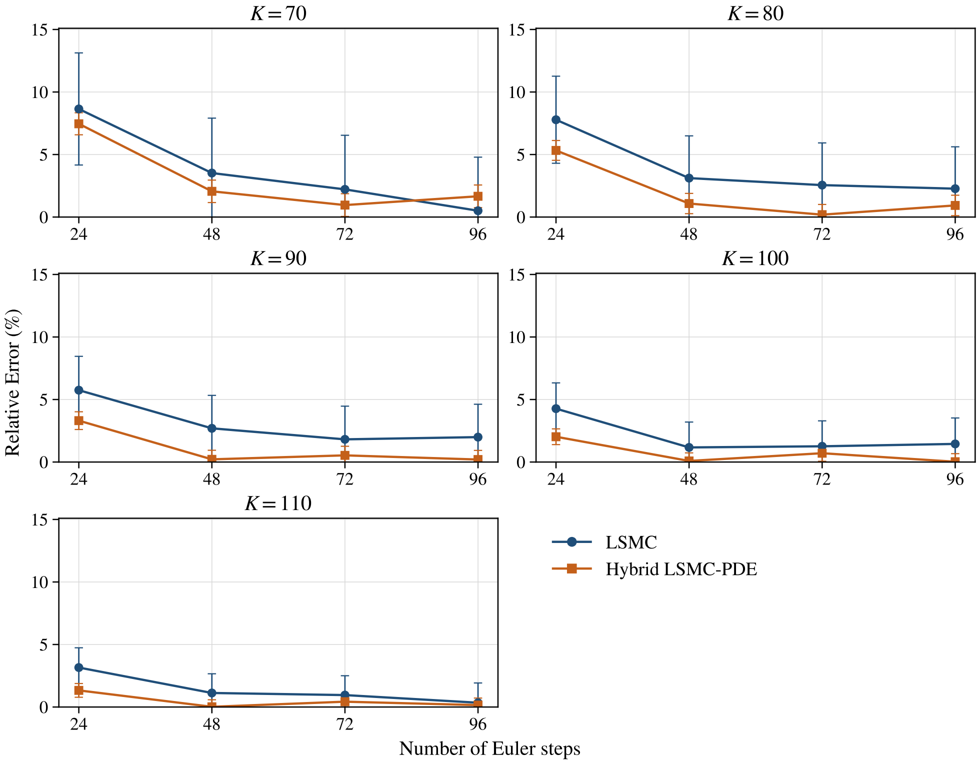

Fig. 1 Reference-relative errors for the matched 20,000-path step-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the estimators.

For 60,000 paths, the same design gives an LSMC reference-error range of 0.102%–7.720% and a Hybrid LSMC–PDE range of 0.028%–7.857%. Figure 2 gives the corresponding errors, and the price grids are reported in Appendix B. The hybrid estimator is not uniformly better at every coarse or fine discretization point, but it gives smaller reference errors in most of the reported configurations, with the clearest differences at the intermediate step levels.

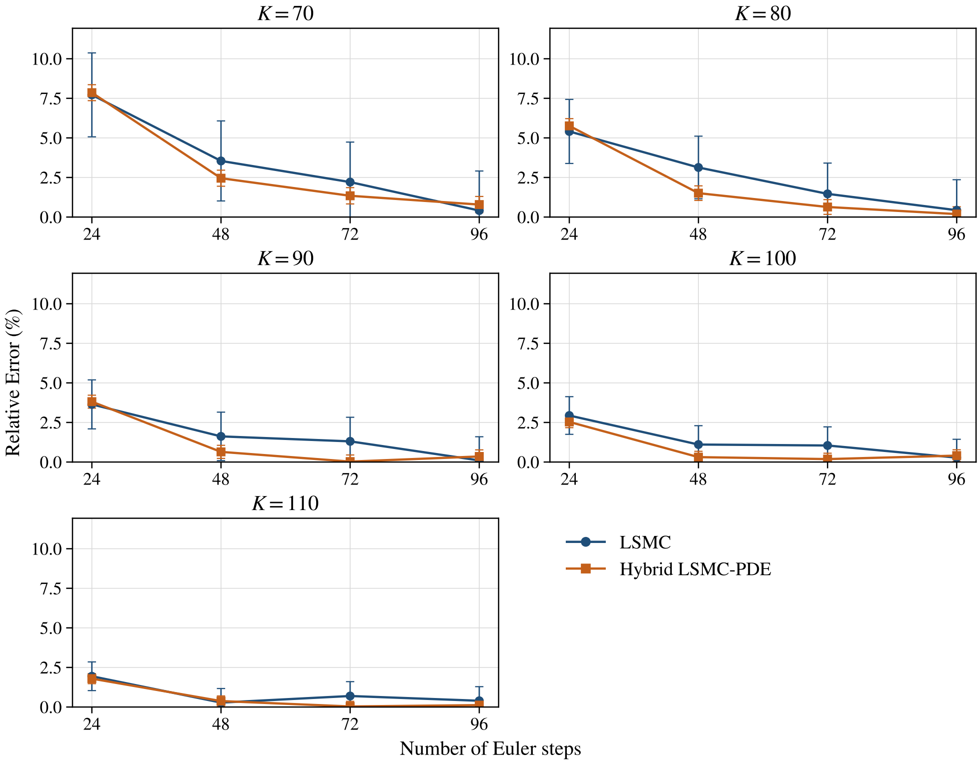

Fig. 2 Reference-relative errors for the matched 60,000-path step-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the estimators.

5.3 Fixed Euler step count and varying path budget

The second experiment fixes the number of Euler time steps and varies the Monte Carlo path budget. The reported path counts are $250, 1000, 5000, 10000, 20000, 40000$, and 60000. Two discretization levels, 48 and 60 Euler steps, are considered. This experiment focuses on sampling variation and regression error.

At 48 Euler steps, the at-the-money strike $K = 100$ gives an LSMC reference-error range of 1.105%–25.948% over the path grid, while the Hybrid LSMC–PDE range is 0.088%–11.118%. Figure 3 shows the path-count sweep for all five strikes. The corresponding pricing estimates are reported in Appendix B.

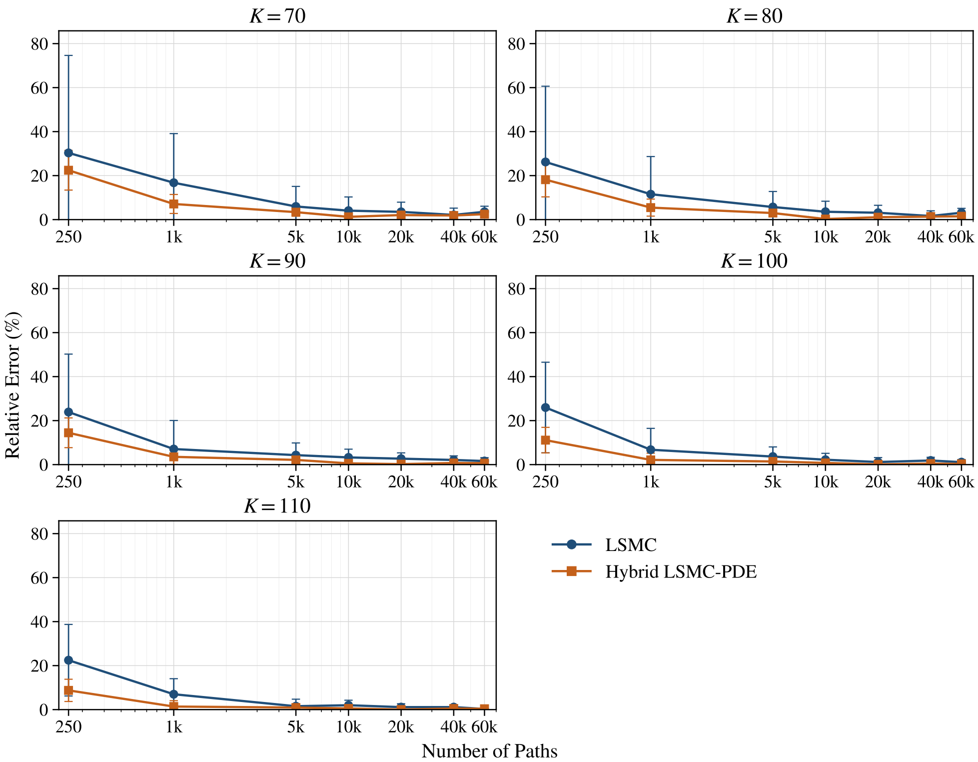

Fig. 3 Reference-relative errors for the fixed 48-step path-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the estimators.

At 60 Euler steps, the corresponding at-the-money LSMC range is 1.324%–28.932%, while the Hybrid LSMC–PDE range is 0.049%–10.102%. Figure 4 shows the same path-count experiment at 60 Euler steps, with the pricing estimates reported in Appendix B. The reduction in reference error is most visible in the low- and moderate-path regimes, where the plain regression estimator is most affected by sampling noise.

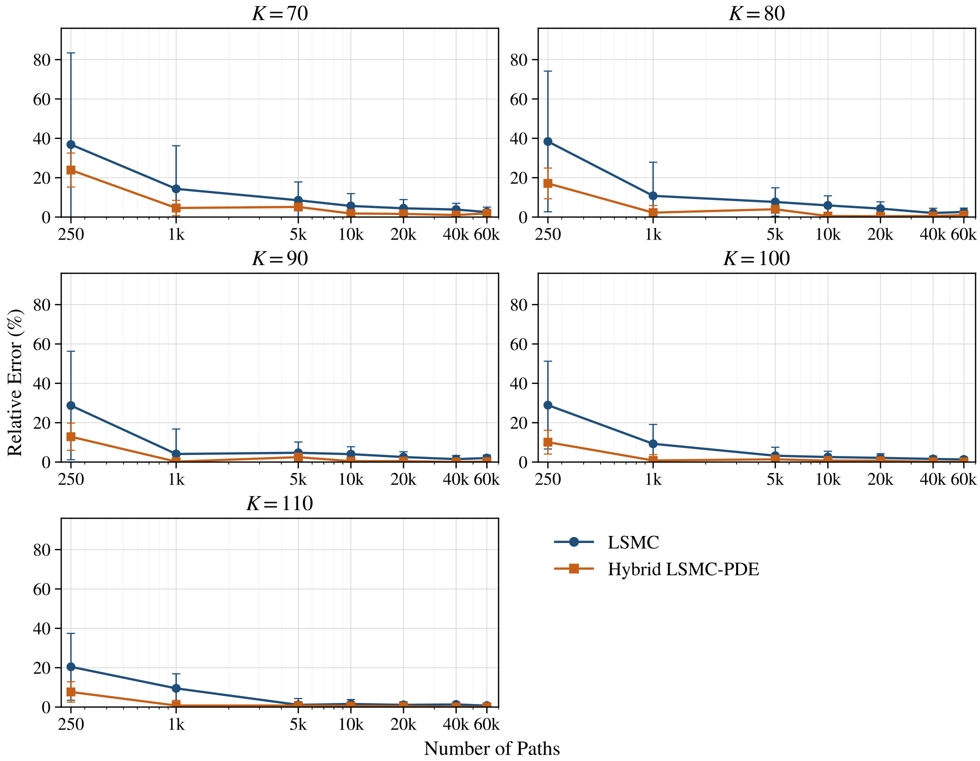

Fig. 4 Reference-relative errors for the fixed 60-step path-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the estimators.

Overall, the experiments are consistent with the conditional PDE step reducing the dimension of the continuation regression. The comparisons are made against the LSMC reference values in Table 2, not exact prices. The full price grids are reported in Appendix B.

6 Conclusion

We studied Bermudan put pricing under a generalized Gatheral double mean-reverting stochastic-volatility model with CEV-type volatility-of-volatility coefficients. The main analytical result is that the generalized variance system is well posed: for $\delta_1, \delta_2 \in [1/2, 1]$, it has a unique strong solution and preserves the nonnegative variance state space.

The hybrid pricing construction uses the one-way coupling of the model. After conditioning on the Brownian increments driving the variance factors and projecting the asset Brownian motion onto these drivers, each conditional asset-price step reduces to a one-dimensional, possibly degenerate, problem in log-price space. The path statistics Z_n, I_n, and B_n are the model-specific quantities needed for the LSMC–PDE method and for the FFT implementation.

The numerical experiments compare the Hybrid LSMC–PDE estimator with plain LSMC under one fixed GDMR parameter set, based on a calibrated double mean-reverting specification from the literature. The hybrid estimator gives smaller reference errors in most of the reported fixed-step and fixed-path configurations, with the clearest gains at low and moderate path budgets. The reference values are LSMC estimates computed with 1200 Euler steps and 1,200,000 paths, not exact prices. Natural extensions include low-biased estimators, broader parameter regimes, and other contract types.

Acknowledgements. The authors thank Anatoliy Malyarenko for helpful discussions.

Declarations

Funding

Not applicable.

Conflict of interest/Competing interests

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The numerical data used to generate the tables and figures are included in the manuscript source files.

Materials availability

Not applicable.

Code availability

Code is available from the corresponding author upon reasonable request.

Author contribution

Mara Kalicanin Dimitrov and Ying Ni contributed to the conceptualization, mathematical analysis, numerical design, interpretation of results, and manuscript preparation. All authors read and approved the final manuscript.

Appendix A Proof of Theorem 2.2

Proof. We work first with the auxiliary equations on $\mathbb{R}$, where the power coefficients are replaced by their positive-part extensions. Once nonnegativity has been proved, the auxiliary equations coincide with the original system on the state space $\mathcal{D}$.

Step 1: construction of v'. Define

$$b_2(x) := \kappa_2(\theta - x), \qquad \sigma_2(x) := \xi_2(x^+)^{\delta_2}, \qquad x \in \mathbb{R}.$$

The drift b_2 is globally Lipschitz and has linear growth. The diffusion coefficient σ_2 is continuous, has at most linear growth, and satisfies the Yamada–Watanabe modulus condition by Assumption 2.1. Therefore pathwise uniqueness holds for

$$\mathrm{d}v'_t = b_2(v'_t)\mathrm{d}t + \sigma_2(v'_t)\mathrm{d}W_t^{(3)}, \qquad v'_0 \geq 0;$$

see [14] and [17, Ch. 5]. Since the coefficients are continuous with linear growth, weak existence holds. Weak existence together with pathwise uniqueness yields a unique strong solution by the Yamada–Watanabe theorem.

It remains to show that the solution does not leave the nonnegative half-line. At the boundary,

$$\sigma_2(0) = 0, \qquad b_2(0) = \kappa_2\theta \geq 0.$$

Moreover, on the negative extension one has

$$\sigma_2(x) = 0, \qquad b_2(x) = \kappa_2(\theta - x) \geq 0, \qquad x \leq 0.$$

Thus the diffusion has no outward noise at the boundary and the drift points inward on the negative side. The one-dimensional stochastic invariance criterion for closed intervals, or equivalently Tanaka's formula applied to the negative part with this positive-part extension, gives

$$v'_t \geq 0, \qquad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 2: construction of v. Having constructed v', define

$$b_1(t,x) := \kappa_1(v'_t - x), \qquad \sigma_1(x) := \xi_1(x^+)^{\delta_1}.$$

For $m \geq 1$, set

$$\tau_m := \inf\{t \geq 0 : v'_t > m\} \wedge T.$$

Since v' has continuous paths on $[0,T]$, we have $\tau_m \uparrow T$ almost surely. On $[0,\tau_m]$, the drift b_1 is progressively measurable, Lipschitz in x with constant κ_1, and satisfies

$$|b_1(t,x)| \leq \kappa_1(m + |x|).$$

The diffusion coefficient σ_1 is continuous, has at most linear growth, and satisfies the Yamada–Watanabe modulus condition. The localized one-dimensional existence and pathwise-uniqueness theorem for SDEs with progressive random coefficients, obtained by the same Yamada–Watanabe argument after localization, therefore gives a unique strong stopped solution of

$$\mathrm{d}v_t = b_1(t,v_t)\mathrm{d}t + \sigma_1(v_t)\mathrm{d}W_t^{(2)}, \qquad t \leq \tau_m.$$

The possible correlation between $W^{(2)}$ and $W^{(3)}$ is immaterial here: pathwise uniqueness compares solutions driven by the same Brownian vector and the same realized input path v'. By pathwise uniqueness, the stopped solutions are compatible on overlaps. Letting $m \to \infty$ gives a unique strong solution for v on $[0,T]$.

The same invariance argument gives nonnegativity. Since $v'_t \geq 0$,

$$b_1(t,0) = \kappa_1 v'_t \geq 0, \qquad \sigma_1(0) = 0.$$

On the negative extension,

$$b_1(t,x) = \kappa_1(v'_t - x) \geq 0, \qquad \sigma_1(x) = 0, \qquad x \leq 0.$$

Hence the closed half-line is invariant and

$$v_t \geq 0, \qquad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 3: construction of S. The process v is continuous and finite on the compact interval $[0,T]$. Therefore

$$\int_0^T v_s \,\mathrm{d}s < \infty \qquad \mathbb{Q}\text{-a.s.}$$

The asset equation is linear in S, and its unique strong solution is

$$S_t = S_0 \exp\left(\int_0^t \left(r - \tfrac{1}{2}v_s\right) \mathrm{d}s + \int_0^t \sqrt{v_s}\, \mathrm{d}W_s^{(1)}\right).$$

Thus $S_t > 0$ for all $t \in [0, T]$ whenever $S_0 > 0$.

The auxiliary positive-part extension is inactive along the constructed variance paths, because $v_t, v_t' \geq 0$. Hence the constructed solution solves the original nonnegative-state system (1). Pathwise uniqueness for the auxiliary triangular system gives pathwise uniqueness for (1).

Finally, existence and pathwise uniqueness for every deterministic initial state imply uniqueness in law. Since the coefficients are time homogeneous, the standard strong Markov property for well-posed time-homogeneous SDEs gives that the solution is a time-homogeneous strong Markov process on $\mathcal{D}$. □

Appendix B Pricing estimates for the numerical experiments

This appendix reports the Bermudan put pricing estimates corresponding to the numerical experiments in Section 5. All entries are rounded to three decimal places. The label LSMC denotes the plain least-squares Monte Carlo estimator, and Hybrid denotes the Hybrid LSMC–PDE estimator.

B.1 Reference prices

Table B1 Reference prices for the parameter set in Table 1. Prices are rounded to three decimal places.

Reference setting	$K = 70$	$K = 80$	$K = 90$	$K = 100$	$K = 110$
1200 steps, 1,200,000 paths	2.523	4.299	6.942	10.682	15.744

Table B2 Pricing estimates in the fixed-path step-sweep experiment with 20,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.741	2.612	2.579	2.536
Hybrid $K = 70$	2.711	2.575	2.547	2.565
LSMC $K = 80$	4.634	4.433	4.409	4.396
Hybrid $K = 80$	4.528	4.345	4.307	4.339
LSMC $K = 90$	7.341	7.129	7.068	7.080
Hybrid $K = 90$	7.172	6.957	6.905	6.956
LSMC $K = 100$	11.138	10.807	10.816	10.837
Hybrid $K = 100$	10.898	10.673	10.607	10.679
LSMC $K = 110$	16.241	15.920	15.894	15.798
Hybrid $K = 110$	15.953	15.747	15.676	15.768

Table B3 Pricing estimates in the fixed-path step-sweep experiment with 60,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.718	2.612	2.579	2.533
Hybrid $K = 70$	2.721	2.585	2.557	2.543
LSMC $K = 80$	4.531	4.434	4.362	4.318
Hybrid $K = 80$	4.547	4.364	4.326	4.307
LSMC $K = 90$	7.195	7.055	7.033	6.949
Hybrid $K = 90$	7.207	6.987	6.940	6.918
LSMC $K = 100$	10.996	10.800	10.794	10.712
Hybrid $K = 100$	10.953	10.715	10.662	10.639
LSMC $K = 110$	16.050	15.788	15.853	15.806
Hybrid $K = 110$	16.026	15.803	15.749	15.725

Table B4 Pricing estimates in the fixed 48-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.288	2.946	2.673	2.625	2.612	2.576	2.612
Hybrid $K = 70$	3.089	2.702	2.608	2.554	2.575	2.570	2.585
LSMC $K = 80$	5.425	4.795	4.545	4.452	4.433	4.369	4.434
Hybrid $K = 80$	5.079	4.533	4.427	4.310	4.345	4.357	4.364
LSMC $K = 90$	8.601	7.433	7.241	7.169	7.129	7.088	7.055
Hybrid $K = 90$	7.946	7.187	7.090	6.903	6.957	6.992	6.987
LSMC $K = 100$	13.454	11.404	11.072	10.918	10.807	10.881	10.800
Hybrid $K = 100$	11.870	10.910	10.833	10.601	10.673	10.730	10.715
LSMC $K = 110$	19.278	16.843	15.985	16.057	15.920	15.921	15.788
Hybrid $K = 110$	17.120	15.962	15.889	15.660	15.747	15.819	15.803

Table B5 Pricing estimates in the fixed 60-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.452	2.884	2.738	2.665	2.635	2.619	2.587
Hybrid $K = 70$	3.125	2.639	2.653	2.568	2.563	2.549	2.570
LSMC $K = 80$	5.950	4.762	4.629	4.553	4.484	4.387	4.411
Hybrid $K = 80$	5.034	4.396	4.470	4.321	4.318	4.321	4.342
LSMC $K = 90$	8.936	7.227	7.270	7.222	7.120	7.045	7.084
Hybrid $K = 90$	7.836	6.957	7.115	6.910	6.910	6.939	6.957
LSMC $K = 100$	13.773	11.673	11.026	10.955	10.910	10.850	10.823
Hybrid $K = 100$	11.761	10.596	10.831	10.604	10.610	10.666	10.677
LSMC $K = 110$	18.962	17.240	15.920	15.991	15.918	15.952	15.860
Hybrid $K = 110$	16.953	15.621	15.863	15.659	15.680	15.755	15.757

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